

Submission Summary

Conference Name	Workshop on Synthetic Data Generation and Management for Building AI Systems
Paper ID	46
Paper Title	A Comparative Study of Transfer Learning Models for Weed and Crop Classification Using Deep Learning
Abstract	Weeds that aren't controlled are still one of the biggest threats to sustainable food production. They not only lower crop yields, but they also make herbicides more expensive, which harms soil health and local ecosystems in ways that last for a long time. It is slow, unreliable, and impossible to do by hand at the scale that modern farming needs to do it. This study examines the efficacy of three lightweight, ImageNet-pretrained architectures MobileNetV1, MobileNetV2, and EfficientNetB0 in automating tasks applied to the V2 Plant Seedlings Dataset, encompassing twelve distinct plant classes. Instead of treating each backbone like a black box, we connect them all to a single classification head made up of Global Average Pooling, Batch Normalisation, and Dropout layers. We then train all of the models in the same two-phase way: 70:20:10 across training, validation, and test partitions, and a suite of augmentation operations is used throughout the training, validation, and test partitions to make the models more robust in real-world situations. For all three networks, we give information on accuracy, precision, recall, F1-score, and confusion-matrix breakdowns. EfficientNetB0 was the clear winner, with a test accuracy of 98.05. MobileNetV1 (87.79) and MobileNetV2 (81.42) were far behind. The results show that transfer learning is a useful and effective way to classify agricultural images. They also give developers who are making weed-management tools for precision farming contexts a clear performance standard to aim for.
Created	2026-05-09 18:16:47 IST
Last Modified	2026-05-09 18:16:47 IST

Authors

Mikhil Sai (Jain University) <23btrcn077@jainuniversity.ac.in>

Srisabari Prahadeshwar G (Jain University) <23btrcn059@jainuniversity.ac.in>

Dr. Suriya Prakash J (Jain University) <Suriyaprakash.j@jainuniversity.ac.in>

Pritesh Jain (Jain University) <23btrcn054@jainuniversity.ac.in>

Gaduputi Kotlo Vijaya (Jain University) <23btrcn061@jainuniversity.ac.in>

Submission Files

A Comparative Study of Transfer Learning Models pcl weed.pdf (564.1 Kb, 2026-05-09 18:13:59 IST)
